THE UNIVERSITY OF ALABAMA IN HUNTSVILLE

ISE 430/530: Manufacturing Systems and Facilities Design

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Industrial & Systems and Engineering Management Department

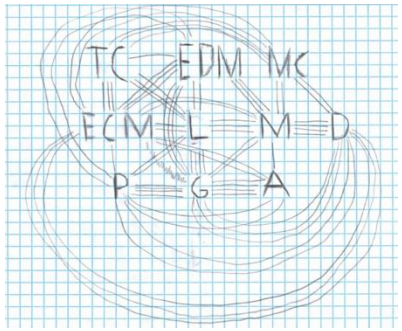
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Project Proposal (Case Competition)

- Please form a group of ~2, and outline the project topic you picked as detailed as possible (you may get some idea from the invited groups from last year).
- I will generate a random order of presentations
- Showcase the from-to chart and the accessibility of data collection from the facility, including detailed information for each department.

Project Proposal

- Each group will present their proposal to the class (time: **3-4 minutes each group**), and we will vote for your most favorable ones other than your own team's project.
- Please use google sheet for the presentation, to save setup time.
- In the vote, indicate the project name, and your team members.
- The final report will be submitted on Nov 20. Please refer to the example of an excellent report.



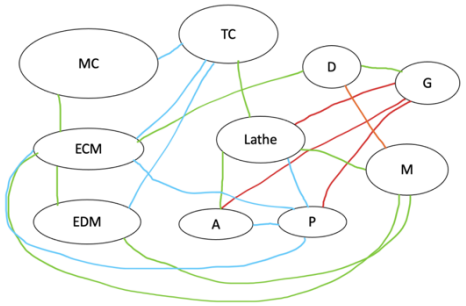
From-To Chart (loads/wk)

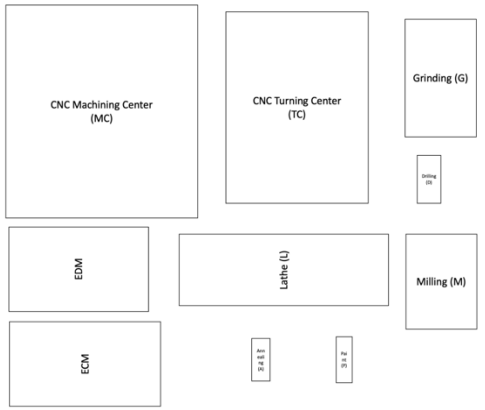
	Milling	Lathes	Grinding	Drilling	EDM	ECM	MC	TC	Anneal	Paint
Milling	-	I	U	E	I	I	U	U	U	U
Lathes		-	A	U	U	U	U	I	U	O
Grinding			-	I	X	X	U	U	A	A
Drilling				-	U	I	U	U	U	U
EDM					-	I	U	O	U	U
ECM						-	I	O	U	O
MC							-	O	U	U
TC								-	U	U
Anneal									-	O
Paint										-

Figure 4. From-To Chart with User Assigned Values.

Priority Codes are assigned to each letter. These numeric values can be used to create a mathematical model. The priority codes chart, shown in Figure 5 can be applied in a value chart, shown in Figure 6







Inadequate drawings

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Project Instruction

Algorithm	Objective Function	Type
Pairwise exchange method	Distance-based	Improvement-based
Graph-based method	Adjacency-based	Construction-based
CRAFT	Distance-based	Improvement-based
BLOCPLAN		
MIP		
LOGIC		
MULTIPLE		

Layout Algorithms

- Pairwise exchange method
- Graph-based method
- CRAFT (Computerized Relative Allocation of Facilities Technique)
- BLOCPLAN
- MIP
- LOGIC (Layout Optimization with Guillotine Induced Cuts)
- MULTIPLE (Multifloor Plant Layout Evaluation)

BLOCPLAN

- Like MicroCRAFT, arranges departments in bands,
- Inputs: relationship chart + a from-to chart,
- Measures layout "cost" using **either the distance-based or adjacency-based objective function**,
- Band number determined by the program; **two or three bands**,
- Band widths may vary,
- Departments contained in only one band - which means they are rectangular,
- Uses **continuous** layout representation,
- Can do both construction and improvement.

BLOCPLAN

- First assigns each department to one of the 2 or 3 bands.
- BLOCPLAN computes the appropriate band width by dividing total area of departments in the band by the building length.
- Note: although it uses both a relationship chart and a from-to chart, the two charts can be used only one at a time while evaluating a layout.

Example 6.4: Adjacency Score

			FLOW							
Department Name	**Area (ft^2)**	**# of Grids**	**A**	**B**	**C**	**D**	**E**	**F**	**G**	**H**
1. A:Receiving	12,000	30	0	45	15	25	10	5	0	0
2 B:Milling	8,000	20	0	0	0	30	25	15	0	0
3 C:Press	6,000	15	0	0	0	0	5	10	0	0
4 D:Screw m/c.	12,000	30	0	20	0	0	35	0	0	0
5 E:Assembly	8,000	20	0	0	0	0	0	65	35	0
6 F:Plating	12,000	30	0	0	0	0	25	0	65	0
7 G:Shipping	12,000	30	0	0	0	0	0	0	0	0
8 H:Dummy	2,000	5	0	0	0	0	0	0	0	0

Department Data and From-to chart

Example 6.4

Flow-Between Chart

	A	B	C	D	E	F	G	H
A	-	45	15	25	10	5	0	0
B		-	0	50	25	20	0	0
C			-	0	5	10	0	0
D				-	35	0	0	0
E					-	90	35	0
F						-	65	0
G							-	0
H								-

Relationship Chart

	A	B	C	D	E	F	G	H
A	-	I	U	O	U	U	U	U
B		-	U	I	O	O	U	U
C			-	U	U	U	U	U
D				-	O	U	U	U
E					-	A	O	U
F						-	E	U
G							-	U
H								-

Add flows from from-to-chart

Assign Relationship Status as A:73-90, E:55-72, I:37-54, O:19-36, U:0-18

Example 6.4

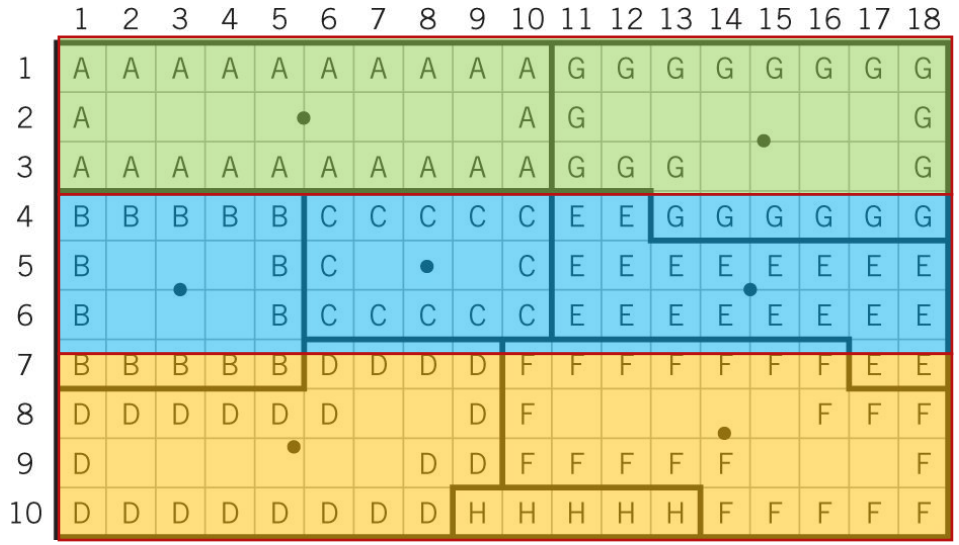
Relationship Chart							
A	**B**	**C**	**D**	**E**	**F**	**G**	**H**
-	I	U	O	U	U	U	U
	-	U	I	O	O	U	U
		-	U	U	U	U	U
			-	O	U	U	U
				-	A	O	U
					-	E	U
						-	U
							-

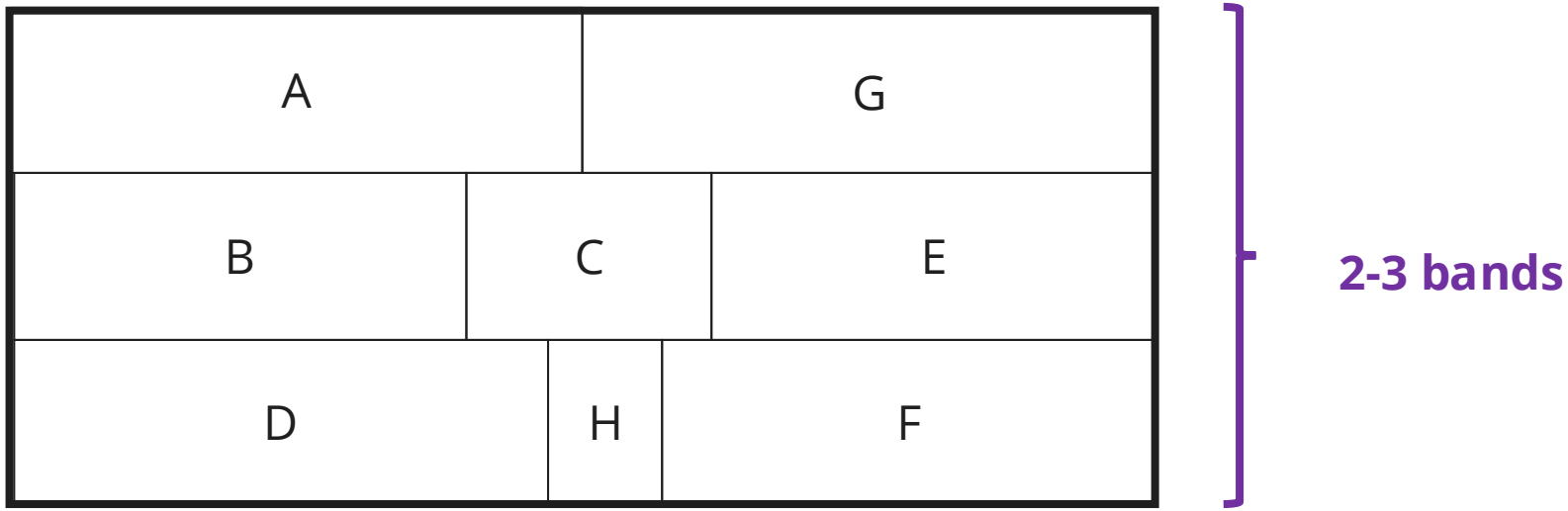
Closeness Rating								
	A	**B**	**C**	**D**	**E**	**F**	**G**	**H**
A	-	2	0	1	0	0	0	0
B		-	0	2	1	1	0	0
C			-	0	0	0	0	0
D				-	1	0	0	0
E					-	10	1	0
F						-	5	0
G							-	0
H								-

Assign Closeness Rating following

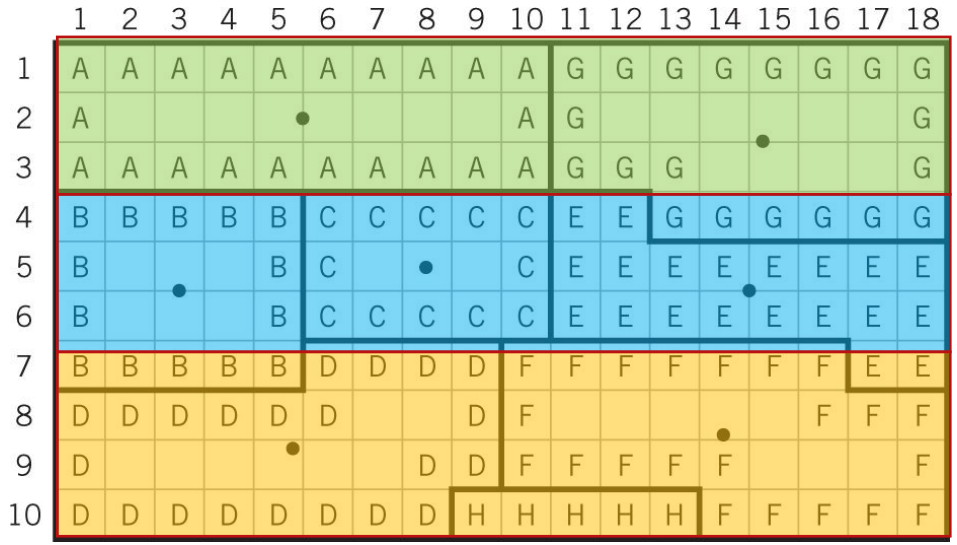
A=10, E=5, I=2, O=1, U=0

Example 6.4

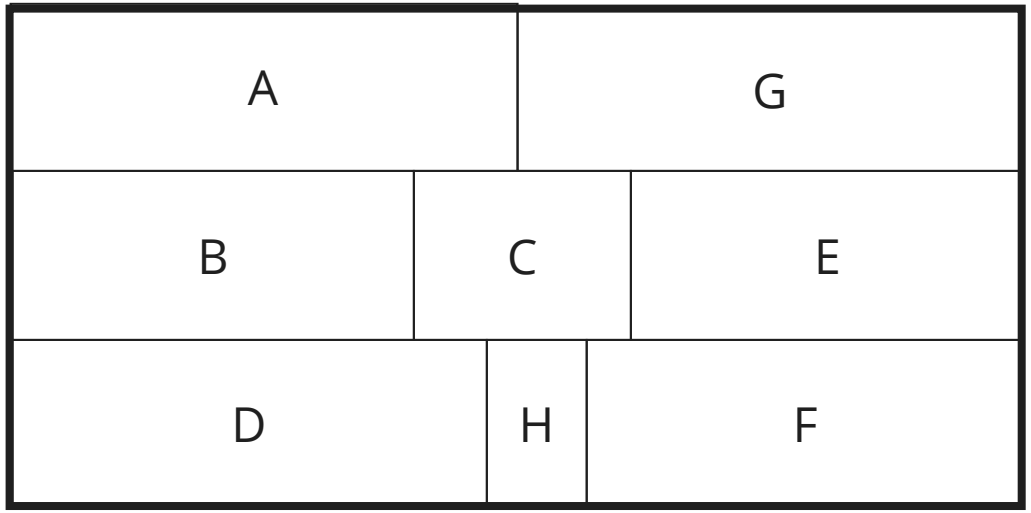




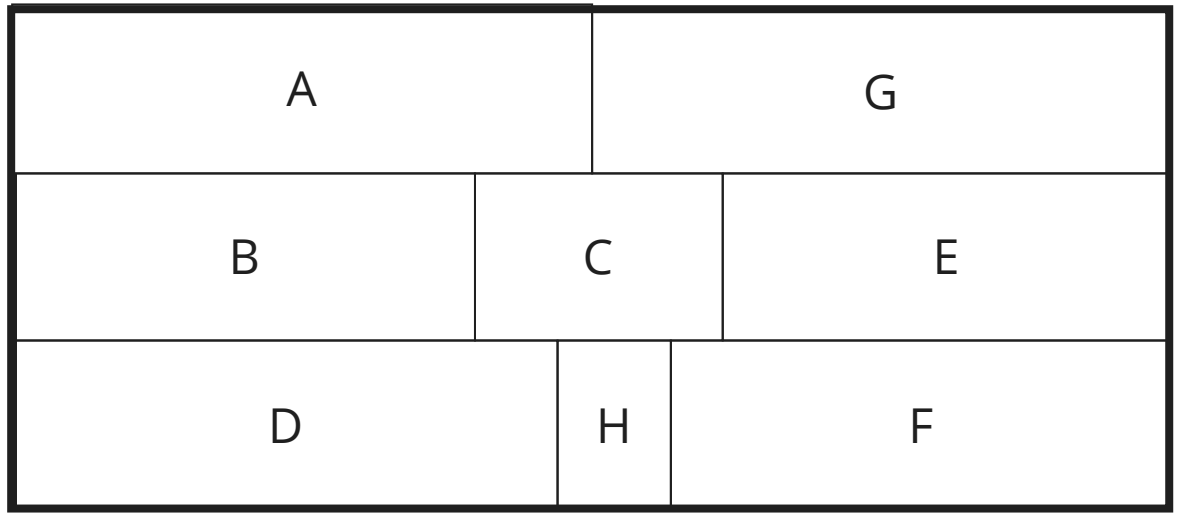
2-3 bands



	# of Grid
A	30
B	20
C	15
D	30
E	20
F	30
G	30
H	5



Example 6.4



Closeness Rating								
	A	**B**	**C**	**D**	**E**	**F**	**G**	**H**
A	-	2	0	1	0	0	0	0
B		-	0	2	1	1	0	0
C			-	0	0	0	0	0
D				-	1	0	0	0
E					-	10	1	0
F						-	5	0
G							-	0
H								-

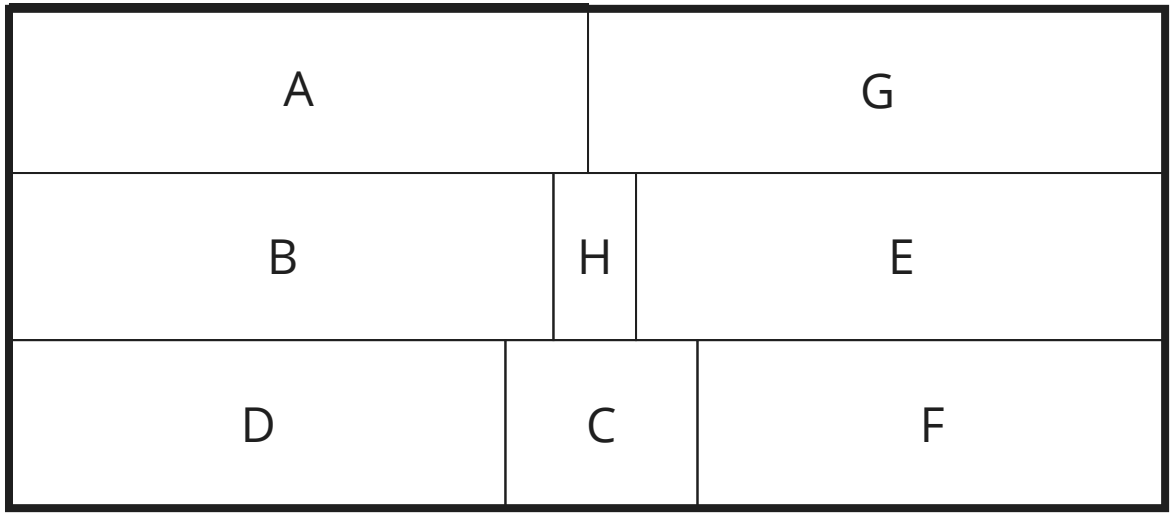
AG, AB, AC, BC, BD, CD, CG, CE, CH, CF, DH, EG, EF, FH

= 0 + 2 + 0 + 0 + 2 + 0 + 0 + 0 + 0 + 0 + 0 + 1 + 10 + 0 = 15

Possible swaps: AB, AC, AD, AE, AF, AG, AH, BC, BD, BE, BF, BG, BH, CD, CE, CF, CG, CH, DE, DF, DG, DH, EF, EG, FG

Based on BLOCPLAN's improvement algorithm (two-way exchanges), exchanging C and H yields the smallest layout cost.

Example 6.4



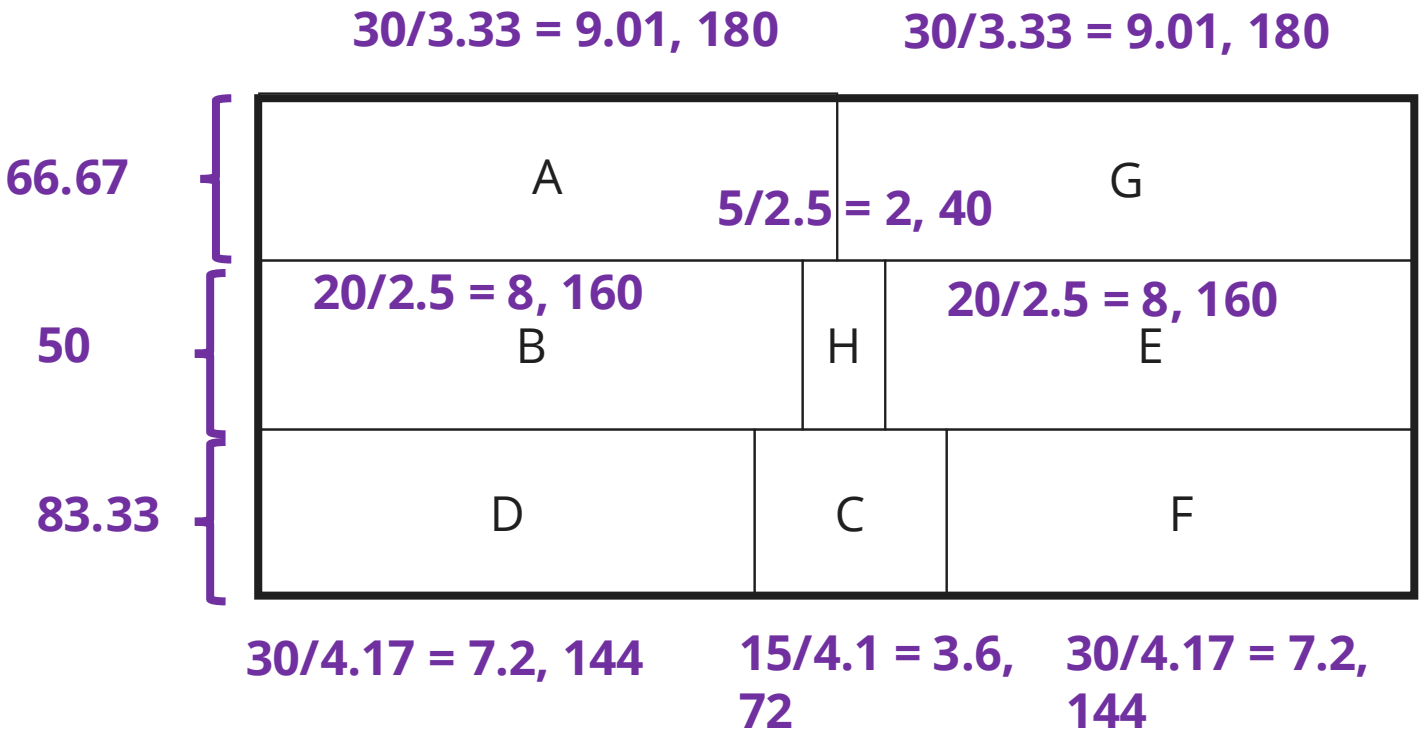
AG, AB, AH, BH, BC, BD, CE, CH, CF, EG, EF, GH

= 0 + 2 + 0 +0 +0 +2 +0 + 0 + 0 +1 + 10 + 0 = 15

Closeness Rating								
	A	**B**	**C**	**D**	**E**	**F**	**G**	**H**
A	-	2	0	1	0	0	0	0
B		-	0	2	1	1	0	0
C			-	0	0	0	0	0
D				-	1	0	0	0
E					-	10	1	0
F						-	5	0
G							-	0
H								-

Example 6.4

	# of Grid
A	30
B	20
C	15
D	30
E	20
F	30
G	30
H	5



A + G = 60

60/18 = 3.33, 66.67 (band width for A and G)

B + H + E = 45

45/18 = 2.5, 50

D + C + F = 75

75/18 = 4.17, 83.33

Layout Algorithms

- Pairwise exchange method
- Graph-based method
- CRAFT (Computerized Relative Allocation of Facilities Technique)
- BLOCPLAN
- MIP
- LOGIC (Layout Optimization with Guillotine Induced Cuts)
- MULTIPLE (Multifloor Plant Layout Evaluation)

Mixed Integer Programming (MIP)

- Continuous representation
- All departments assumed to be rectangular
- Uses distance-based objective function
  - Rectilinear distance between centroids
- Construction algorithm
- Can be used to determine machine placement

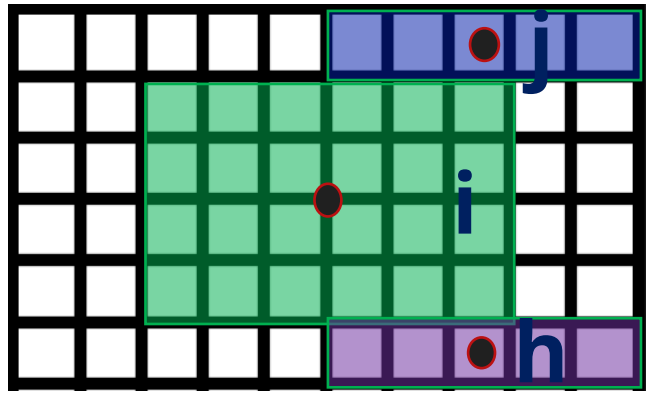
MIP-Parameters

- B_x=building length
- B_y=building width
- A_i=area of department *i*
- L_i^l=lower limit on the length of department *i*
- L_i^u=upper limit on the length of department *i*
- W_i^l=lower limit on the width of department *i*
- W_i^u=upper limit on the width of department *I*

Decision Variables

- ❿ α_i = *x*-coordinate of the centroid of department *i*
- ❿ β_i = y-coordinate of the centroid of department *i*
- x'_i = *x*-coordinate of the left (or west) side of department *i*
- x''_i = *x*-coordinate of the right (or east) side of department *i*
- y'_i = *y*-coordinate of the bottom (or south) side of department *i*
- y''_i = *y*-coordinate of the top (or north) side of department *i*
- z^x_{ij}= equal to 1 if department *i* is strictly to the east of department *j*
- z^y_{ij}= equal to 1 if department *i* is strictly to the north of department *j*

MIP-Parameters



α_i = *x*-coordinate of the centroid of department *i*
β_i = y-coordinate of the centroid of department *i*
x'_i = *x*-coordinate of the left (or west) side of department *i*
x''_i = *x*-coordinate of the right (or east) side of department *i*
y'_i = *y*-coordinate of the bottom (or south) side of department *i*
y''_i = *y*-coordinate of the top (or north) side of department *i*
z^x_{ij}= equal to 1 if department *i* is strictly to the east of department *j*
z^y_{ij}= equal to 1 if department *i* is strictly to the north of department *j*

$$\alpha_i = 5, \beta_i = 3$$

$$x'_i = 2, x''_i = 8$$

$$y'_i = 1, y''_i = 5$$

$$\alpha_j = 7.5,$$

$$\beta_j = 5.5$$

$$\alpha_h = 7.5, \beta_h = 0.5$$

$$z^y_{ij} = 0, z^y_{ji} = 1, z^x_{ij} = 0, z^x_{ji} = 0$$

$$z^x_{ih} = 0, z^y_{ih} = 1 \; z^x_{hi} = 0, z^y_{hi} = 0$$

$$z^y_{ij} + z^x_{ij} + z^y_{ji} + z^x_{ji} \geq 1,$$
$$\forall i, j, i \neq j$$

For any two departments, one is at least north, south, east, west of the other department: there is no overlap of departments

MIP-Model

$$z = \sum_i \sum_j f_{ij} c_{ij} \left(|\alpha_i - \alpha_j| + |\beta_i - \beta_j| \right) \quad \boldsymbol{d_{ij}}$$

subject to :

$L_i^l \le (x_i'' - x_i') \le L_i^u \quad$ for all i — Length restriction for department

$W_i^l \le (y_i'' - y_i') \le W_i^u \quad$ for all i — Width restriction for department

$(x_i'' - x_i')(y_i'' - y_i') = A_i \quad$ for all i — Space Requirement

$0 \le x_i' \le x_i'' \le B_x \quad$ for all i

$0 \le y_i' \le y_i'' \le B_y \quad$ for all i — Building/lot size restriction

$\alpha_i = 0.5x_i' + 0.5x'' \quad$ for all i

$\beta_i = 0.5y_i' + 0.5y_i'' \quad$ for all i — Centroid coordinates

If $z_{ij}^x = 1 \Rightarrow x_j'' \le x_i' \quad$ for all i and $j, i \ne j$

If $z_{ij}^y = 1 \Rightarrow y_j'' \le y_i' \quad$ for all i and $j, i \ne j$

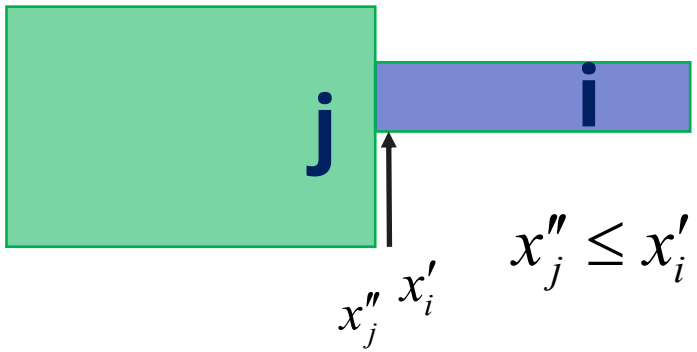
$z_{ij}^x + z_{ji}^x + z_{ij}^y + z_{ji}^y \ge 1 \quad$ for all i and $j, i < j$

$\alpha_i, \beta_i \ge 0 \quad$ for all i

$x_i', x_i'', y_i', y_i'' \ge 0 \quad$ for all i

$z_{ij}^x, z_{ij}^y \, 0/1 \quad$ for all i and $j, i \ne j$

MIP-Model



$$z = \sum_i \sum_j f_{ij} c_{ij} \left(|\alpha_i - \alpha_j| + |\beta_i - \beta_j| \right) \; \boldsymbol{d_{ij}}$$

subject to :

$$L_i^l \le (x_i'' - x_i') \le L_i^u \quad \text{for all } i$$

Length restriction for department

$$W_i^l \le (y_i'' - y_i') \le W_i^u \quad \text{for all } i$$

Width restriction for department

$$(x_i'' - x_i')(y_i'' - y_i') = A_i \quad \text{for all } i$$

Space Requirement

$$0 \le x_i' \le x_i'' \le B_x \quad \text{for all } i$$

$$0 \le y_i' \le y_i'' \le B_y \quad \text{for all } i$$

Building/lot size restriction

$$\alpha_i = 0.5x_i' + 0.5x'' \quad \text{for all } i$$

$$\beta_i = 0.5y_i' + 0.5y_i'' \quad \text{for all } i$$

Centroid coordinates

$$\text{If } z_{ij}^x = 1 \Rightarrow x_j'' \le x_i' \quad \text{for all } i \text{ and } j, i \ne j$$

$$\text{If } z_{ij}^y = 1 \Rightarrow y_j'' \le y_i' \quad \text{for all } i \text{ and } j, i \ne j$$

If department i is east of department j, right wall x coordinate of j department must be less than or equal to left wall x coordinate of i

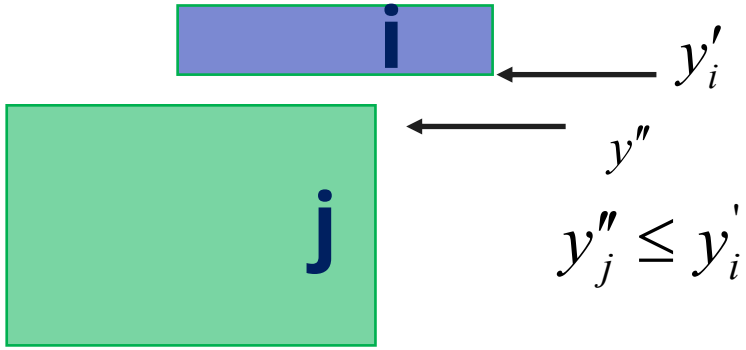
$$z_{ij}^x + z_{ji}^x + z_{ij}^y + z_{ji}^y \ge 1 \quad \text{for all } i \text{ and } j, i < j$$

$$\alpha_i, \beta_i \ge 0 \quad \text{for all } i$$

$$x_i', x_i'', y_i', y_i'' \ge 0 \quad \text{for all } i$$

$$z_{ij}^x, z_{ij}^y \; 0/1 \quad \text{for all } i \text{ and } j, i \ne j$$

MIP-Model



$$z = \sum_i \sum_j f_{ij} c_{ij} \left(|\alpha_i - \alpha_j| + |\beta_i - \beta_j| \right) \quad d_{ij}$$

subject to :

$L_i^l \le (x_i'' - x_i') \le L_i^u$ for all i — Length restriction for department

$W_i^l \le (y_i'' - y_i') \le W_i^u$ for all i — Width restriction for department

$(x_i'' - x_i')(y_i'' - y_i') = A_i$ for all i — Space Requirement

$0 \le x_i' \le x_i'' \le B_x$ for all i

$0 \le y_i' \le y_i'' \le B_y$ for all i — Building/lot size restriction

$\alpha_i = 0.5x_i' + 0.5x''$ for all i

$\beta_i = 0.5y_i' + 0.5y_i''$ for all i — Centroid coordinates

If $z_{ij}^x = 1 \Rightarrow x_j'' \le x_i'$ for all i and $j, i \ne j$

If $z_{ij}^y = 1 \Rightarrow y_j'' \le y_i'$ for all i and $j, i \ne j$

$z_{ij}^x + z_{ji}^x + z_{ij}^y + z_{ji}^y \ge 1$ for all i and $j, i < j$

$\alpha_i, \beta_i \ge 0$ for all i

$x_i', x_i'', y_i', y_i'' \ge 0$ for all i

$z_{ij}^x, z_{ij}^y 0/1$ for all i and $j, i \ne j$

If department i is north of department j, upper wall y coordinate of j department must be less than or equal to bottom wall y coordinate of i

MIP-Model

$$z = \sum_i \sum_j f_{ij} c_{ij} \left(\left| \alpha_i - \alpha_j \right| + \left| \beta_i - \beta_j \right| \right) \; \boldsymbol{d_{ij}}$$

subject to :

$$L_i^l \le (x_i'' - x_i') \le L_i^u \quad \text{for all } i$$

Length restriction for department

$$W_i^l \le (y_i'' - y_i') \le W_i^u \quad \text{for all } i$$

Width restriction for department

$$(x_i'' - x_i')(y_i'' - y_i') = A_i \quad \text{for all } i$$

Space Requirement

$$0 \le x_i' \le x_i'' \le B_x \quad \text{for all } i$$

$$0 \le y_i' \le y_i'' \le B_y \quad \text{for all } i$$

Building/lot size restriction

$$\alpha_i = 0.5x_i' + 0.5x'' \quad \text{for all } i$$

$$\beta_i = 0.5y_i' + 0.5y_i'' \quad \text{for all } i$$

Centroid coordinates

$$\text{If } z_{ij}^x = 1 \Rightarrow x_j'' \le x_i' \quad \text{for all } i \text{ and } j, i \ne j$$

$$\text{If } z_{ij}^y = 1 \Rightarrow y_j'' \le y_i' \quad \text{for all } i \text{ and } j, i \ne j$$

$$z_{ij}^x + z_{ji}^x + z_{ij}^y + z_{ji}^y \ge 1 \quad \text{for all } i \text{ and } j, i < j$$

For any two departments, one is at least north, south, east, west of the other department: there is no overlap of departments

$$\alpha_i, \beta_i \ge 0 \quad \text{for all } i$$

$$x_i', x_i'', y_i', y_i'' \ge 0 \quad \text{for all } i$$

Non negativity constraints

$$z_{ij}^x, z_{ij}^y \; 0/1 \quad \text{for all } i \text{ and } j, i \ne j$$